

LABORATORY OBSERVATION/RECORD

Course:	Full Stack Web Development
Course Code:	23CM4121
Experiment No.:	Task 5
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1. TITLE

Interactive Product Price Calculator Using HTML, CSS and JavaScript

2. AIM

To create a webpage containing a product name, quantity and price, and to calculate the total price dynamically using JavaScript functions and event listeners whenever the quantity or price changes, displaying the result without refreshing the page.

3. OBJECTIVE

The objectives of this experiment are:

- To design an input form using HTML labels and input fields.
- To style the page as a card layout using an external CSS file.
- To access page elements using the DOM (`getElementById`).
- To write reusable JavaScript functions for validation and calculation.
- To use event listeners so the result updates as the user types.
- To format and display the result without reloading the page.

4. THEORY

HTML defines the structure of the page: the input fields for product name, quantity and unit price, and the result area. **CSS** controls the appearance such as the card, spacing and colours. **JavaScript** reads the input values and updates the result.

- **DOM** represents the webpage as objects that JavaScript can read and change.
- **getElementById()** selects an element using its id.
- **addEventListener()** runs a function when an event such as *input* occurs on an element.
- **input event** fires on every change of a field, so the total updates while the user types.
- **parseFloat()** converts the entered text into a decimal number; **isNaN()** detects invalid or empty values.
- **Math.floor()** keeps the quantity a whole number.
- **toFixed(2)** displays amounts with two decimal places.
- **forEach()** attaches the same listener to several fields in one loop.
- **textContent** updates the displayed text of an element.

Flow: User input → input event → `updateTotal()` → validate values → total = quantity × price → result updated

5. ALGORITHM / PROCEDURE

1. Create a folder and add the files calculator.html, style.css and script.js.
2. In the HTML file, create labelled inputs for product name, quantity and price, and a result area.
3. Link style.css in the head and script.js at the end of the body.
4. Style the page with CSS (card layout, input boxes and highlighted result).
5. In JavaScript, store references to the input fields and result elements.
6. Define `readNumber ( )` to convert a field value into a valid non-negative number (0 if empty or invalid).
7. Define `updateTotal ( )` to read the quantity and price, calculate $\text{total} = \text{quantity} \times \text{price}$ and show it with two decimals.
8. Use `forEach ( )` to attach an `input` listener to all three fields.
9. Call `updateTotal ( )` once at load to show the initial result.
10. Open calculator.html in a browser and enter different values.
11. Perform multiple test cases and verify the actual output against the expected output.

6. PROGRAM

calculator.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Product Price Calculator</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="card">
    <h1>Product Price Calculator</h1>

    <div class="row">
      <label for="pname">Product Name</label>
      <input type="text" id="pname" placeholder="e.g. Laptop">
    </div>

    <div class="row">
      <label for="qty">Quantity</label>
      <input type="number" id="qty" value="1" min="1" step="1">
    </div>

    <div class="row">
      <label for="unitPrice">Price per unit (Rs.)</label>
      <input type="number" id="unitPrice" value="0" min="0" step="0.01">
    </div>

    <div class="result">
      <p id="summary">Product: 1 x Rs. 0.00</p>
      <p id="grandTotal">Total: Rs. 0.00</p>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

script.js

```
// Form fields and result elements
```

```

const nameField = document.getElementById('pname');
const qtyField = document.getElementById('qty');
const priceField = document.getElementById('unitPrice');
const summaryLine = document.getElementById('summary');
const totalLine = document.getElementById('grandTotal');

// Convert a field value to a safe non-negative number
function readNumber(field) {
  const value = parseFloat(field.value);
  if (isNaN(value) || value < 0) {
    return 0;
  }
  return value;
}

// Recalculate and show the total price
function updateTotal() {
  const quantity = Math.floor(readNumber(qtyField));
  const price = readNumber(priceField);
  const total = quantity * price;
  const product = nameField.value.trim() || 'Product';

  summaryLine.textContent = product + ': ' + quantity + ' x Rs. ' + price.toFixed(2);
  totalLine.textContent = 'Total: Rs. ' + total.toFixed(2);
}

// Update the result whenever any field changes
[nameField, qtyField, priceField].forEach(function (field) {
  field.addEventListener('input', updateTotal);
});

// Show the initial result on page load
updateTotal();

```

style.css

```

body {
  font-family: Verdana, Geneva, sans-serif;
  background: #eef2f7;
  margin: 0;
  padding: 30px;
}

.card {
  width: 380px;
  margin: 0 auto;
  padding: 22px 26px;
  background: #ffffff;
  border: 1px solid #c8d1dc;
  border-radius: 10px;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.12);
}

h1 {
  font-size: 20px;
  text-align: center;
  color: #1f3a5f;
  margin: 0 0 18px 0;
}

.row {
  margin-bottom: 14px;
}

label {
  display: block;

```

```
    font-size: 13px;
    margin-bottom: 4px;
    color: #333333;
}

input {
    width: 100%;
    padding: 7px;
    font-size: 14px;
    border: 1px solid #9aa7b5;
    border-radius: 4px;
    box-sizing: border-box;
}

.result {
    margin-top: 18px;
    padding: 10px 14px;
    background: #e8f5e9;
    border-left: 5px solid #2e7d32;
    border-radius: 4px;
}

#summary {
    margin: 0 0 6px 0;
    font-size: 13px;
    color: #444444;
}

#grandTotal {
    margin: 0;
    font-size: 19px;
    font-weight: bold;
    color: #1b5e20;
}
```

7. CODE EXPLANATION

Code	Explanation
<code>getElementById('pname')</code>	Selects the product name input field.
<code>getElementById('qty')</code>	Selects the quantity input field.
<code>getElementById('unitPrice')</code>	Selects the price input field.
<code>readNumber(field)</code>	Converts a field value to a number; returns 0 if empty, invalid or negative.
<code>Math.floor(...)</code>	Rounds the quantity down to a whole number.
<code>updateTotal()</code>	Calculates quantity $\times$ price and updates the summary and total lines.
<code>nameField.value.trim()</code>	Removes extra spaces; 'Product' is used if the name is empty.
<code>toFixed(2)</code>	Shows the price and total with two decimal places.
<code>textContent</code>	Replaces the text of the result elements.
<code>[...].forEach(function...)</code>	Adds the input listener to every field.
<code>addEventListener('input', ...)</code>	Runs <code>updateTotal()</code> on every change.
<code>updateTotal();</code>	Displays the initial result on page load.

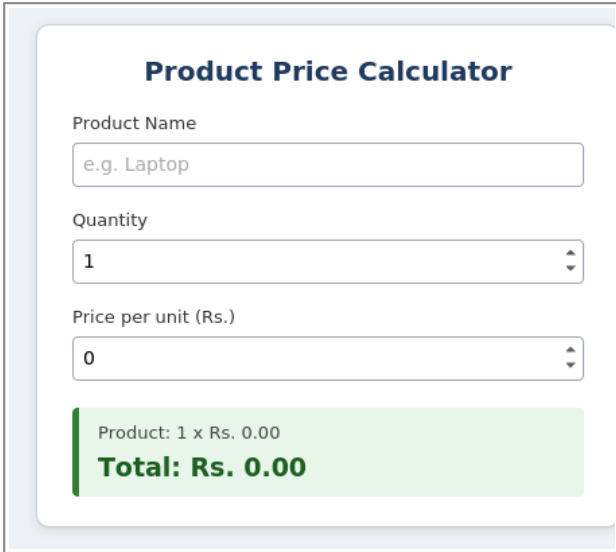
8. TEST CASES AND EXECUTION DATA

Each test case was executed in the browser by entering the values below and observing the result shown on the page. The results were recorded exactly as displayed.

Test Case	Product	Qty	Price	Expected Result	Actual Result	Status
TC-01	Laptop	2	500	Total: Rs. 1000.00	Total: Rs. 1000.00	Pass
TC-02	Notebook	12	45.50	Total: Rs. 546.00	Total: Rs. 546.00	Pass
TC-03	(empty)	1 (default)	0 (default)	Total: Rs. 0.00	Total: Rs. 0.00	Pass
TC-04a	Headphones	3	1299.99	Total: Rs. 3899.97	Total: Rs. 3899.97	Pass
TC-04b	Headphones	3 $\rightarrow$ 5 (changed)	1299.99	Total updates to Rs. 6499.95 without reload	Total: Rs. 6499.95 (no reload)	Pass
TC-05	Keyboard	2	Left empty	Total: Rs. 0.00 (empty treated as 0)	Total: Rs. 0.00	Pass
TC-06	Mouse	4	-100	Total: Rs. 0.00 (negative treated as 0)	Total: Rs. 0.00	Pass

9. OUTPUT

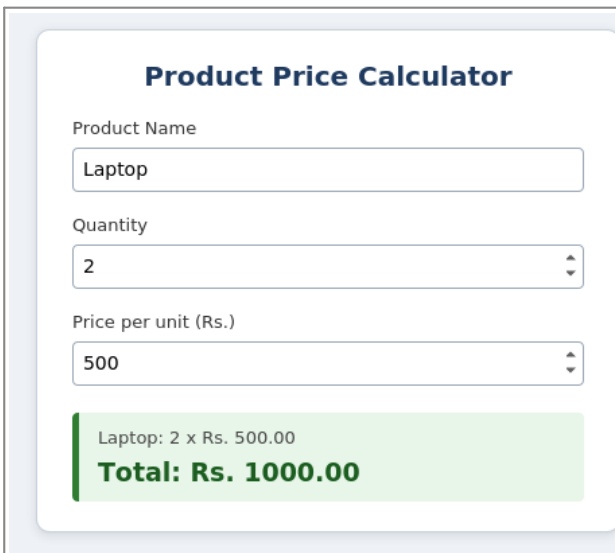
Output 1: Initial page load



The screenshot shows a web form titled "Product Price Calculator". It has three input fields: "Product Name" with the placeholder text "e.g. Laptop", "Quantity" with the value "1", and "Price per unit (Rs.)" with the value "0". Below these fields, a green box displays the calculation: "Product: 1 x Rs. 0.00" and "Total: Rs. 0.00".

On load the page shows **Total: Rs. 0.00** (quantity 1 × price 0).

Output 2: TC-01 – Laptop, Quantity 2, Price 500



The screenshot shows the same "Product Price Calculator" form, but with updated values. The "Product Name" field now contains "Laptop", the "Quantity" field contains "2", and the "Price per unit (Rs.)" field contains "500". The green box at the bottom now displays: "Laptop: 2 x Rs. 500.00" and "Total: Rs. 1000.00".

Total: Rs. 1000.00 displayed.

Output 3: TC-02 – Notebook, Quantity 12, Price 45.50

Product Price Calculator

Product Name

Quantity

Price per unit (Rs.)

Notebook: 12 x Rs. 45.50

Total: Rs. 546.00

Total: Rs. 546.00 displayed (decimal price handled).

Output 4: TC-03 – Empty name, default quantity and price

Product Price Calculator

Product Name

Quantity

Price per unit (Rs.)

Product: 1 x Rs. 0.00

Total: Rs. 0.00

Total: Rs. 0.00 displayed; “Product” is used as the default name.

Output 5: TC-04a – Headphones, Quantity 3, Price 1299.99

Product Price Calculator

Product Name

Quantity

Price per unit (Rs.)

Headphones: 3 x Rs. 1299.99
Total: Rs. 3899.97

Total: Rs. 3899.97 displayed.

Output 6: TC-04b – Quantity changed from 3 to 5 (same page, no refresh)

Product Price Calculator

Product Name

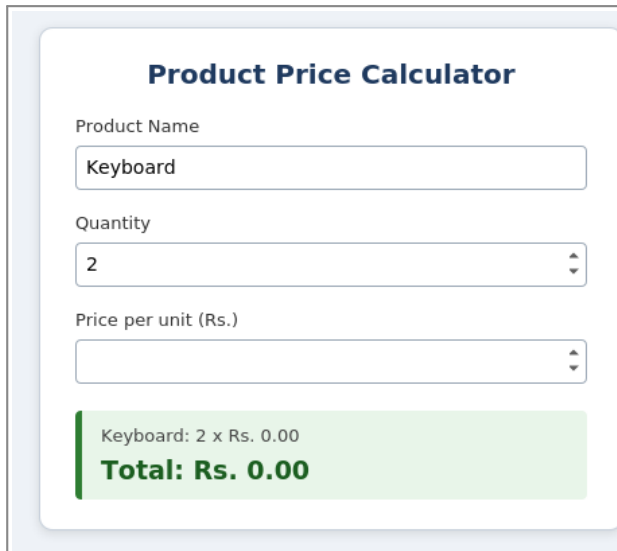
Quantity

Price per unit (Rs.)

Headphones: 5 x Rs. 1299.99
Total: Rs. 6499.95

The total updated instantly to Rs. 6499.95 as soon as the quantity changed.

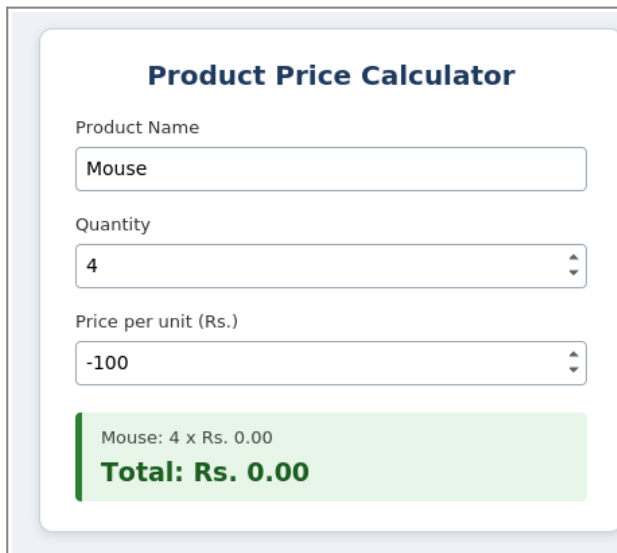
Output 7: TC-05 – Keyboard, Quantity 2, Price left empty



The screenshot shows a web form titled "Product Price Calculator". It has three input fields: "Product Name" with the value "Keyboard", "Quantity" with the value "2", and "Price per unit (Rs.)" which is empty. Below the inputs, a green box displays the calculation: "Keyboard: 2 x Rs. 0.00" and the total "Total: Rs. 0.00".

Total: Rs. 0.00 displayed; the empty price is treated as 0 and no error occurs.

Output 8: TC-06 – Mouse, Quantity 4, Price -100



The screenshot shows the same "Product Price Calculator" form. The "Product Name" is "Mouse", "Quantity" is "4", and "Price per unit (Rs.)" is "-100". The green result box shows: "Mouse: 4 x Rs. 0.00" and the total "Total: Rs. 0.00".

Total: Rs. 0.00 displayed; the negative price is treated as 0.

10. RESULT

Thus, an interactive Product Price Calculator was successfully developed using HTML, CSS and JavaScript. JavaScript functions and event listeners were used to calculate the total price dynamically whenever the quantity or price changed, and the result was displayed without refreshing the page. The application was verified using multiple test cases.